

Econ 101A

Section 6 (Solutions)

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This week's lecture put the consumer problem to work. We solve it for the CES family, we look at the cases where the tangency recipe misleads you, and we introduce the **indirect utility function**. The midterm leans hard on all three, so treat the exercises as exam rehearsal, not as new theory.

A note on the sign convention. The lecture defines the marginal rate of substitution as the *slope* of an indifference curve with x on the horizontal axis,

$$\text{MRS} = \frac{dy}{dx} = -\frac{u_x}{u_y} < 0.$$

The budget line has slope $-p/q$. Tangency means the two slopes are equal, which is the same as $u_x/u_y = p/q$. On the exam, keep the minus sign when you are asked for the MRS itself; the exam grades the sign.

1 The CES Family and Its Special Cases

The lecture worked with the CES utility

$$u(x, y) = (\alpha x^\rho + \beta y^\rho)^{1/\rho}.$$

One functional form nests the three shapes you already know, indexed by ρ :

- $\rho \rightarrow 0$: **Cobb–Douglas**, $u \propto x^\alpha y^\beta$. Constant budget shares.
- $\rho = 1$: **perfect substitutes**, $u = \alpha x + \beta y$. Straight indifference lines, generically a corner.
- $\rho \rightarrow -\infty$: **perfect complements** (Leontief), $u = \min\{x, y\}$. L-shaped, a kink.

The sign of ρ tells you the substitution pattern: $\rho > 0$ the goods are (net) substitutes, $\rho < 0$ they are (net) complements. Keep $\rho < 1$ so preferences are convex and the tangency you find is a maximum. That last clause is the bridge to the next section.

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2 When the Tangency Recipe Misleads You

The recipe “set MRS equal to the price ratio and solve with the budget” finds the right answer only when the optimum is interior and preferences are convex. Three cases break it, and each shows up somewhere on the midterm.

1. Non-convex preferences (the tangency is a minimum). If preferences are concave to the origin (for CES this is $\rho > 1$), the interior stationary point the FOCs hand you is the *worst* affordable bundle, and the maximum is a corner. The tell is the second-order condition: a bordered Hessian with $\det \bar{H} < 0$, or simply that utility is higher at a corner than at the tangency point. This is exactly the short-question flavor of $u = x^2 + y^2$: the FOCs point to the middle of the budget line, which is the closest point to the origin, i.e. the minimum.

2. Corner solutions from a baseline good. When one good enters with an additive baseline, e.g. $u = x(y + a)$, low income drives y^* to zero and the tangency formula returns a negative y^* , which is nonsense. You then set $y = 0$ and spend everything on x . Always check that the interior bundle you computed is actually nonnegative.

3. Multiplicity. With perfect substitutes and a price ratio exactly equal to the marginal-utility ratio ($p/q = \alpha/\beta$), every bundle on the budget line is optimal. Off that knife-edge you get a unique corner.

3 The Indirect Utility Function

Plug the Marshallian demands back into utility:

$$v(p, q, M) = u(x^*(p, q, M), y^*(p, q, M)).$$

This is the best utility attainable at those prices and income. Three properties, all from the lecture:

- Increasing in M : a bigger budget set cannot lower the maximum.
- Decreasing in each price p_i : a smaller budget set cannot raise it.
- Homogeneous of degree zero in (p, q, M) : scaling all prices and income together leaves the budget set, hence the optimum, unchanged.

The key comparative static comes from the envelope theorem applied to the Lagrangian $\mathcal{L} = u + \lambda(M - px - qy)$:

$$\frac{\partial v}{\partial M} = \frac{\partial \mathcal{L}}{\partial M} \Big|_{\text{opt}} = \lambda, \quad \frac{\partial v}{\partial p} = \frac{\partial \mathcal{L}}{\partial p} \Big|_{\text{opt}} = -\lambda x^*.$$

So λ , the multiplier from Section 3, is the **marginal utility of income**: one extra dollar raises maximized utility by about λ . (Dividing the two gives $x^* = -(\partial v / \partial p) / (\partial v / \partial M)$, which is Roy's identity; you will see it named next week.)

4 Exercises

Exercise 1 (the full pipeline on Cobb–Douglas)

Let $u(x, y) = \sqrt{xy}$ with prices $p, q > 0$ and income $M > 0$.

- Compute the marginal utilities and the MRS (with its sign).
- Set up the Lagrangian, write the first-order conditions, and solve for the Marshallian demands $x^*(p, q, M)$ and $y^*(p, q, M)$.
- Find the indirect utility $v(p, q, M)$ and the multiplier λ . Verify that $\partial v / \partial M = \lambda$.
- Without redoing the algebra, state the demands for $u = x^{1/3}y^{2/3}$.

Solution. (a) $u_x = \frac{1}{2}\sqrt{y/x}$, $u_y = \frac{1}{2}\sqrt{x/y}$, so $\text{MRS} = -u_x/u_y = -y/x$.

(b) $\mathcal{L} = \sqrt{xy} + \lambda(M - px - qy)$. FOCs: $\frac{1}{2}\sqrt{y/x} = \lambda p$, $\frac{1}{2}\sqrt{x/y} = \lambda q$, and $px + qy = M$. Dividing the first two, $y/x = p/q$, i.e. $qy = px$: the two goods get equal expenditure shares. With the budget, $px = qy = M/2$, so

$$x^* = \frac{M}{2p}, \quad y^* = \frac{M}{2q}.$$

(c) $v = \sqrt{x^*y^*} = \sqrt{\frac{M}{2p} \frac{M}{2q}} = \frac{M}{2\sqrt{pq}}$. From the first FOC, $\lambda = \frac{u_x}{p} = \frac{1}{2p} \sqrt{y^*/x^*} = \frac{1}{2p} \sqrt{p/q} = \frac{1}{2\sqrt{pq}}$.

And $\frac{\partial v}{\partial M} = \frac{1}{2\sqrt{pq}} = \lambda$. ✓

(d) Cobb–Douglas spends constant budget shares equal to the exponents: $x^* = \frac{1}{3} \frac{M}{p}$, $y^* = \frac{2}{3} \frac{M}{q}$. (Taking $\sqrt{\cdot}$ or any increasing transform does not change demand; see Exercise 3.)

Exercise 2 (a corner solution)

A consumer has $u(x, y) = x(y + 5)$ with prices $p, q > 0$ and income M .

- Assume an interior solution and use tangency plus the budget to find x^* and y^* .
- For which incomes is that interior solution valid? What goes wrong below the threshold?
- Give the optimal bundle when income is below the threshold.

Solution. (a) $u_x = y + 5$, $u_y = x$, so tangency $u_x/u_y = (y + 5)/x = p/q$, i.e. $x = \frac{q(y + 5)}{p}$.

Substitute into $px + qy = M$: $q(y + 5) + qy = M$, so $q(2y + 5) = M$ and

$$y^* = \frac{M}{2q} - \frac{5}{2}, \quad x^* = \frac{M + 5q}{2p}.$$

(b) Valid only if $y^* \geq 0$, i.e. $M \geq 5q$. Below that, the formula returns a negative y^* , which is infeasible: the consumer's baseline satisfaction from y (the “+5”) makes x so attractive that at low income she wants no y at all.

(c) For $M < 5q$ set $y^* = 0$ and spend everything on x : $x^* = M/p$, with utility $u = 5M/p$. (Check: at $y = 0$, $\text{MRS magnitude} = 5/(M/p) = 5p/M \geq p/q$ exactly when $M \leq 5q$, confirming the corner.)

Exercise 3 (transforms and perfect substitutes)

- (a) A consumer's preferences are represented by $u(x, y)$. Show that maximizing $v = e^u$ subject to the same budget yields the *same* demands. What about $w = -12u$? Explain the general principle.
- (b) Now let $u(x, y) = 2x + y$ (perfect substitutes) with prices $p, q > 0$ and income M . Find the optimal bundle as a function of prices, and identify the price ratio at which the solution is not unique.

Solution. (a) Any strictly increasing transform g gives $g(u(a)) \geq g(u(b)) \iff u(a) \geq u(b)$, so it ranks bundles identically and the maximizer is unchanged; e^u is strictly increasing, so demands are identical. $w = -12u$ is strictly *decreasing*: it reverses every ranking, so maximizing w would minimize u ; it does *not* represent the same preferences. The principle: demand depends only on the ordering of bundles, i.e. on preferences, not on the particular utility numbers.

(b) MRS magnitude $= u_x/u_y = 2$, constant. Compare to p/q :

$$\text{if } 2 > \frac{p}{q} \ (p < 2q) : x^* = \frac{M}{p}, y^* = 0; \quad \text{if } 2 < \frac{p}{q} \ (p > 2q) : x^* = 0, y^* = \frac{M}{q}.$$

Buy only the good with the higher “bang per buck.” At $p/q = 2$ the indifference lines are parallel to the budget line and *any* bundle on the budget line is optimal.